ADVANCED PROBLEMS FOR LECTURE 18

In the lecture, we will be talking at a more general level of the patterns of what is going on. But there are deeper questions to explore therein, which we cannot get to. This is your opportunity to investigate deeper for yourself. Don't limit yourself to just these questions, let your curiosity guide you too! That is, if you wonder about any details, or if any certain patterns or ideas exist, see if you can state your question clearly, and answer it! Here are a few suggestions to get you started.

(1) Prove that if f is continuous on $[a, b]$ and $f(x) \geq 0$ for all $x \in [a, b]$, then

$$\int_a^b f(t)dt = 0 \quad \text{implies} \quad f = 0$$

(2) We saw in the previous problem that if a function is nonnegative and continuous on an interval, then it must have positive integral (area) if it is positive at any point. But you need continuity to prove this. Can you describe all functions f which are not continuous, but are nonnegative on $[a, b]$ such that

$$\int_a^b f(t)dt = 0?$$

(3) Look back to the advanced problems for chapter 15 for definitions of linear maps (or google it). Let V be the space of continuous functions on $[0, 1]$. Let W be the space of functions which are continuous on $[0, 1]$ and differentiable on $(0, 1)$
   (a) For any f in V, let $Int : V \to W$ be defined by

$$Int(f)(x) = \int_0^x f(t)dt$$

   (b) Show that Int is a right-inverse for the differentiation operator D (see advanced lecture 15), but that it is not a left-inverse. (It turns out to be something called a pseudo-inverse.)
   (c) Describe the kernel of Int.

(4) Can we invert the quotient rule? Try coming up with your own integration rule based on the quotient rule, and try to write down a pattern of what to look for so that it can be applied.

(5) Given that the inverted rules for differentiation can look messy, we might try to come up with our own nicer rules. But we can see almost immediately that we run into a brick wall. List a few products of functions that do not behave nicely under anti-differentiation, thus tossing away our hopes of a nice "rule".